Example 8.21

If $\vec{a} = -3\hat{i} + 4\hat{j} - 7\hat{k}$ and $\vec{b} = 6\hat{i} + 2\hat{j} - 3\hat{k}$,

verify (i) $\vec{a}$ and $\vec{a} \times \vec{b}$ are perpendicular to each other.

(ii) $\vec{b}$ and $\vec{a} \times \vec{b}$ are perpendicular to each other.

Solution :

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 4 & -7 \\ 6 & 2 & -3 \end{vmatrix} = \hat{i}(-12+14) - \hat{j}(9+42) + \hat{k}(-6-24) = 2\hat{i} - 51\hat{j} - 30\hat{k}$$

(i) $\vec{a} \cdot (\vec{a} \times \vec{b}) = (-3\hat{i} + 4\hat{j} - 7\hat{k}) \cdot (2\hat{i} - 51\hat{j} - 30\hat{k}) = -6 - 204 + 210 = 0$.

Therefore, $\vec{a}$ and $\vec{a} \times \vec{b}$ are perpendicular.

(ii) $\vec{b} \cdot (\vec{a} \times \vec{b}) = (6\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (2\hat{i} - 51\hat{j} - 30\hat{k}) = 12 - 102 + 90 = 0.$

Therefore $\vec{b}$ and $\vec{a} \times \vec{b}$ are perpendicular.

Example 8.22

Find the vectors of magnitude 6 which are perpendicular to both vectors

$\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{i} + j - 2\hat{k}$.

Solution

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} = \hat{i}(2-3) - \hat{j}(-8+6) + \hat{k}(4-2) = -\hat{i} + 2\hat{j} + 2\hat{k}$$

Unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ are $\pm \dfrac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|} = \pm \left[\dfrac{-\hat{i} + 2\hat{j} + 2\hat{k}}{3} \right]$

Vectors of magnitude 6 perpendicular to both $\vec{a}$ and $\vec{b}$ are $\pm 2(-\hat{i} + 2\hat{j} + 2\hat{k})$.

Example 8.23

Find the cosine and sine angle between the vectors $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = 4\hat{i} - 2\hat{j} + 2\hat{k}$.

Solution

Let θ be the angle between $\vec{a}$ and $\vec{b}$

$$\vec{a} \cdot \vec{b} = (2\hat{i} + \hat{j} + 3\hat{k}) \cdot (4\hat{i} - 2\hat{j} + 2\hat{k}) = 8 - 2 + 6 = 12$$

$$|\vec{a}| = |2\hat{i} + \hat{j} + 3\hat{k}| = \sqrt{14} \quad ; \quad |\vec{b}| = |4\hat{i} - 2\hat{j} + 2\hat{k}| = \sqrt{24}$$

$$\cos\theta = \frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} = \frac{12}{\sqrt{14}\sqrt{24}} = \sqrt{\frac{3}{7}}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 3 \\ 4 & -2 & 2 \end{vmatrix} = \hat{i}(2+6) - \hat{j}(4-12) + \hat{k}(-4-4) = 8\hat{i} + 8\hat{j} - 8\hat{k}$$

$$|\vec{a} \times \vec{b}| = |8\hat{i} + 8\hat{j} - 8\hat{k}| = 8\sqrt{3}$$

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \frac{8\sqrt{3}}{\sqrt{14}\sqrt{24}} = \frac{4\sqrt{3}}{\sqrt{7}\sqrt{12}} = \frac{2}{\sqrt{7}}.$$

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